Frobenius generators for smooth toric varieties

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Abstract. On any smooth complex toric variety, a line bundle has a toric Frobenius direct image generating the derived category if and only if its restriction to every linear toric inclusion has nonzero total cohomology. This proves a conjecture of Hanlon, Hicks and Lazarev, and shows that whenever generation is possible, every sufficiently large integer exponent works, without a completeness or projectivity assumption. The same criterion leads to a finite test in the original fan and an explicit successful exponent. We also prove that the sets of line-bundle summands are eventually periodic, with a period depending only on the rays and a threshold linear in the divisor coefficient height.

1. Introduction

Toric Frobenius turns a single line bundle into a finite collection of line bundles. Hanlon, Hicks and Lazarev proved that the collection obtained from the structure sheaf generates the derived category of every smooth toric variety [6]. For an arbitrary line bundle, they identified cohomological obstructions on linear toric subvarieties and conjectured that these are the only obstructions.

We prove this conjecture for every smooth complex toric variety and show that generation, once possible, holds for every sufficiently large integer exponent. Both the criterion and a successful exponent can be computed from the original fan and the divisor coefficients.

Let $F_\ell : X \to X$ be the finite toric map induced by multiplication by ℓ on the lattice of one-parameter subgroups. Its direct image of a line bundle is a sum of line bundles; generation will mean classical generation of $D^b(\operatorname{Coh} X)$.

Write $X = X_\Sigma$, where Σ is a finite smooth fan in $N_\mathbb{R}$, and put $M = N^\vee$. A rational subspace $W \subset N_\mathbb{R}$ is *linear for* Σ if $\sigma \cap W$ belongs to Σ for every $\sigma \in \Sigma$. The restricted fan in the saturated lattice $N \cap W$ then defines a closed toric inclusion

$$i_W : Y_W = X_{\Sigma|W, N\cap W} \hookrightarrow X.$$

We call these the *linear inclusions*. The cohomological condition below includes all degrees.

Theorem 1.1. *Let D be a Cartier divisor on a smooth complex toric variety X. A direct image $F_{\ell *}\mathcal{O}_X(D)$ classically generates $D^b(\operatorname{Coh} X)$ for some $\ell \geq 1$ if and only if*

$$R\Gamma(Y_W, \mathcal{O}_X(D)|_{Y_W}) \neq 0 \qquad \text{for every linear inclusion } Y_W \hookrightarrow X.$$

Whenever this condition holds, $F_{\ell *}\mathcal{O}_X(D)$ generates for every sufficiently large integer ℓ. No completeness or projectivity assumption is required.

For example, the only linear inclusions in $\mathbb{P}^n$ are the identity and the identity point of its dense torus. The criterion therefore says that a Frobenius direct image of $\mathcal{O}_{\mathbb{P}^n}(d)$ generates precisely when

$$d \geq 0 \qquad \text{or} \qquad d \leq -n-1.$$

These are exactly the line bundles with nonzero total cohomology; see also [6]. The criterion concerns cohomology in all degrees, rather than positivity of the line bundle.

The theorem reduces generation of an entire derived category to cohomology on a distinguished family of subvarieties. Necessity follows from adjunction, while sufficiency uses the coherent–constructible correspondence to locate the obstruction geometrically. On the real dual torus, the normal space of a maximal support stratum determines a linear inclusion whose nonzero cohomology supplies a compact test. Torus covers shrink this test into the stratum, where it detects the given nonzero object. The shrinking works for all sufficiently large exponents, and a Cartier-boundary compactification extends the argument from complete fans to arbitrary smooth fans.

The effective results require no search over all power maps. Cohomology of finitely many induced simplicial complexes and integer linear feasibility decide the criterion. A common denominator of the fan arrangement gives a successful exponent. We also prove eventual periodicity of the sets of direct summands, with a period depending only on the rays and a threshold bounded linearly in the divisor coefficient height. The appendix constructs the sheaf model from the cellular resolutions of [6].

Conventions. We work over $k = \mathbb{C}$. Derived Hom denotes the full derived Hom complex, and all sheaf functors are derived. Large sheaf categories allow unbounded complexes with arbitrary-dimensional stalk cohomology. We write $D_{\mathrm{qc}}(X)$ for the unbounded derived category of quasi-coherent complexes. Since X is smooth, $\operatorname{Perf}(X) = D^b(\operatorname{Coh} X)$. Every line bundle has an invariant divisor representative; trivialise it on the dense torus, whose Laurent coordinate ring is factorial, and take the divisor of the resulting rational section.

2. **The linear obstruction**

The necessity can be seen directly from a linear inclusion.

Lemma 2.1. *If $i : Y \hookrightarrow X$ is a linear inclusion of codimension c, then*

$$\mathrm{RHom}_X(i_*\mathcal{O}_Y, F_{\ell *}\mathcal{O}_X(D)) \simeq R\Gamma(Y, \mathcal{O}_X(D)|_Y)^{\oplus \ell^c}[-c].$$

Proof. The inclusion has trivial normal bundle. Indeed, on the toric open set whose cones lie in W, a lattice splitting identifies it with $Y \times \{1\} \subset Y \times (\mathbb{C}^*)^c$. The inverse image of Y under F_ℓ is the disjoint union of ℓ^c torus translates of Y, and torus translation preserves the invariant line bundle. Flat pullback, adjunction, and the regular-embedding duality formula therefore identify the left side with the displayed sum of equal complexes. □

If $F_{\ell *}\mathcal{O}_X(D)$ generates and the displayed complex vanishes, then $i_*\mathcal{O}_Y$ is left orthogonal to every perfect complex. It is itself perfect and nonzero, a contradiction. This proves the necessity in Theorem 1.1.

3. **Generation for complete fans**

Throughout this section Σ is smooth and complete, of dimension n. Put $T_M = M_{\mathbb{R}}/M$ and let $p : M_{\mathbb{R}} \to T_M$ be the quotient map.

We will detect every nonzero object by a morphism to the proposed generator. The sheaf model turns this into a local problem; choose a stratum of largest dimension on which the object is nonzero, and fit a suitable test sheaf into a small neighborhood of that stratum. The linear inclusion in the theorem arises from its normal space.

3.1. **The sheaf model**

We use the coherent–constructible correspondence in the normalisation of [7, (9.14)]. It gives a colimit-preserving equivalence

$$\kappa_\Sigma : D_{\mathrm{qc}}(X) \xrightarrow{\sim} \mathrm{Sh}^{\diamond}_{\Lambda_\Sigma}(T_M), \qquad \Lambda_\Sigma = \bigcup_{\sigma \in \Sigma} p(\sigma^\perp) \times (-\sigma). \tag{1}$$

The compact objects correspond to bounded finite constructible complexes, and $\kappa_\Sigma(\mathcal{O}_X) = k_{\{0\}}$ [7, Theorem 6.2, Lemma 10.1, Corollary 12.10]. An equivariant structure on $\mathcal{O}_X(D)$ supplies a bounded finite polyhedrally constructible complex K on $M_{\mathbb{R}}$, with compact support, such that

$$\kappa_\Sigma(\mathcal{O}_X(D)) \simeq p_! K \simeq p_* K.$$

This follows from the equivariant correspondence and its compatibility with forgetting equivariance [4, Theorem 7.1], [10, Theorem 2.4]. Here the equality of proper and ordinary direct image uses the compact support of K, not properness of p.

If an injective lattice map defines a toric map $g : X_1 \to X_2$ and the inverse image of each cone is a union of cones, its dual real-torus map $q : T_{M_2} \to T_{M_1}$ satisfies

$$\kappa_1 L g^* \simeq q_! \kappa_2 \tag{2}$$

[7, Proposition 9.4]. This applies to power maps and linear inclusions. These statements also follow from the cellular construction in Appendix A.

3.2. **The geometry of a support stratum**

The periodic hyperplanes

$$H_{\rho,a} = \{x \in M_{\mathbb{R}} : \langle x, v_\rho \rangle = a\}, \qquad \rho \in \Sigma(1), \quad a \in \mathbb{Z},$$

induce a finite stratification $\mathcal{S}$ of T_M by connected, relatively open polyhedra. A full-dimensional smooth cone supplies a lattice basis among the rays, so each stratum lifts injectively to a bounded rational polytope. This is a Whitney stratification, and Λ_Σ lies in its conormal union. Consequently every object of the category in (1) is weakly constructible for $\mathcal{S}$, also with unbounded coefficients [5, Proposition 4.8].

For a stratum S, let $V_S \subset M_{\mathbb{R}}$ be its tangent space and put $W_S = V_S^{\perp}$.

Maximality of the support stratum makes the complex locally supported on a closed slice; all incident strata of larger dimension carry zero stalks. Its microsupport consequently contains every normal direction to that slice, and the toric microsupport condition forces those directions to be covered by a complete subfan. This subfan is the linear inclusion needed below.

Lemma 3.1. *Suppose $F \neq 0$ belongs to $\mathrm{Sh}^{\diamond}_{\Lambda_\Sigma}(T_M)$, and S has maximal dimension among its nonzero strata. Then W_S defines a complete linear inclusion, and $S \subset p(V_S)$.*

Proof. Near a point $x \in S$, choose a lift $\widetilde{x}$ and a ball B meeting no hyperplane that does not contain the local lift of S. Every other stratum meeting B has larger dimension. Thus

$$p^{-1}F|_B \simeq i_* \underline{E}_{(\widetilde{x}+V_S)\cap B}$$

for a nonzero coefficient complex E. The microsupport at x contains the whole normal space W_S.

Every cone contributing to the fibre $(\Lambda_\Sigma)_x$ is contained in W_S; its ray hyperplanes contain S. These cones cover W_S, so they form a complete subfan there. Given

$\sigma \in \Sigma$, choose $v \in \operatorname{relint}(\sigma \cap W_S)$ and a cone $\eta \subset W_S$ containing v. The face $\sigma \cap \eta$ of σ contains v. Every face containing this relative-interior point contains $\sigma \cap W_S$, whence $\sigma \cap W_S = \sigma \cap \eta$. Thus W_S is linear for Σ.

One of the contributing cones has dimension $\dim W_S$. For that cone η, we have $\eta^\perp = V_S$ and $x \in p(\eta^\perp) = p(V_S)$. The same argument applies throughout S, proving the last assertion. □

Retain the finitely many strata for which $W_S \subset (\Lambda_\Sigma)_x$ for $x \in S$. This condition is constant on a stratum. The proof shows that each retained W_S defines a complete linear inclusion, each retained stratum lies in $p(V_S)$, and every maximal support stratum is retained.

3.3. **Compact tests**

Nonvanishing on a linear inclusion must now be converted into a test that can be moved and shrunk. On the dual torus, the inclusion corresponds to a quotient by a subtorus. Lifting that subtorus to $M_\mathbb{R}$ separates the cohomology into tests on parallel affine planes; compact support ensures that only finitely many occur.

Let $Y = Y_W$ be a complete linear inclusion and set $V = W^\perp$. The dual map $q : T_M \to T_{M_Y}$ has connected kernel $T_V = p(V)$. By (2), properness of q, and adjunction,

$$R\Gamma(Y, \mathcal{O}_X(D)|_Y) \simeq \mathrm{RHom}_{T_M}(k_{T_V}, p_*K). \tag{3}$$

Pulling back by p decomposes k_{T_V} over the locally finite family of planes $V + m$, $m \in M/(M \cap V)$. Only finitely many meet $\operatorname{supp} K$. Hence nonvanishing in (3) gives an $m \in M$ with

$$A_{V,m} := \mathrm{RHom}_{M_\mathbb{R}}(k_{V+m}, K) \neq 0. \tag{4}$$

This complex is bounded and finite-dimensional, since it is the cohomology of the compactly supported finite constructible complex $j_m^! K$ on $V + m$.

For $u \in M$, put $a_{\ell,u}(z) = \ell z - u$ and $K_{\ell,u} = a_{\ell,u}^{-1} K$. If $f_\ell : T_M \to T_M$ is multiplication by ℓ, the covering-space fibre product gives

$$f_\ell^! p_* K \simeq \bigoplus_{u \in M/\ell M} p_* K_{\ell,u}, \qquad \operatorname{supp} K_{\ell,u} = \frac{u + \operatorname{supp} K}{\ell}. \tag{5}$$

Replacing u by $u + \ell m$ translates the lifted sheaf by m and does not change its direct image. For every F in the category (1), coherent adjunction and (2) give

$$\mathrm{RHom}(F, \kappa_\Sigma(F_{\ell *}\mathcal{O}_X(D))) \simeq \mathrm{RHom}(F, f_\ell^! p_* K). \tag{6}$$

The right side is computed in the ambient sheaf category; its target need not have microsupport in Λ_Σ.

3.4. **From compact tests to generation**

For smooth proper X, a perfect complex G generates if and only if its left orthogonal in $D_{\mathrm{qc}}(X)$ vanishes. Indeed, with $S = (-)\otimes\omega_X[n]$, Serre duality identifies $\mathrm{RHom}_X(F,G)$ with $\mathrm{RHom}_X(S^{-1}G,F)^\vee$, first for perfect F and then by filtered homotopy colimits for all F. Vector-space duality detects zero, so compact generation [9, Tag 09SR] identifies this condition with classical generation by $S^{-1}G$. Apply the autoequivalence S.

Proposition 3.2. *If the restriction cohomology is nonzero on every complete linear inclusion, then $F_{\ell*}\mathcal{O}_X(D)$ generates for every sufficiently large integer ℓ.*

Proof. First choose the tests independently of the object to be detected. For each retained stratum choose a lift $x_S \in V_S$ of a point of S, a ball B_S as in Lemma 3.1, and a witness m_S for (4). Put $\Lambda_S = M \cap V_S$. Choose a bounded fundamental parallelepiped Q_S for this lattice. For every ℓ there is $v_{\ell,S} \in \Lambda_S$ with $q_{\ell,S} = \ell x_S - v_{\ell,S} \in Q_S$. Thus a lattice point approximates ℓx_S with bounded error, which tends to zero after division by ℓ without imposing any divisibility condition on the exponent. Taking $u_{\ell,S} = v_{\ell,S} - m_S$ gives

$$a_{\ell,u_{\ell,S}}(V_S) = V_S + m_S, \qquad \operatorname{supp} K_{\ell,u_{\ell,S}} = x_S + \frac{\operatorname{supp} K - m_S - q_{\ell,S}}{\ell}.$$

The last support is compactly contained in B_S for all sufficiently large ℓ, uniformly over the finite set of retained strata.

Now fix an exponent beyond this common threshold, let $F \neq 0$, and choose a maximal support stratum S. On B_S the source is the constant complex $E \neq 0$ on the closed slice V_S. Because the target is compactly supported inside this ball, adjunction allows the Hom calculation to be made entirely there. The affine change of variables then gives

$$\mathrm{RHom}_{T_M}(F, p_* K_{\ell,u_{\ell,S}}) \simeq \mathrm{RHom}_{M_{\mathbb{R}}}(\underline{E}_{V_S}, K_{\ell,u_{\ell,S}}) \simeq \mathrm{RHom}_k(E, A_{V_S,m_S}) \neq 0.$$

The last nonvanishing holds even for unbounded E; complexes of vector spaces split into their cohomology, and nonzero vector spaces admit nonzero maps to any nonzero vector space. By (5) and (6), every nonzero F therefore pairs nontrivially with $\kappa_\Sigma(F_{\ell*}\mathcal{O}_X(D))$. The left-orthogonal criterion proves generation. □

4. **Removing completeness**

The complete case supplies the desired generators after restriction, provided the line bundle extends with nonzero cohomology on every complete linear inclusion. An effective Cartier boundary allows us to arrange this simultaneously; twisting by its

multiples does not change the bundle on X, and its cohomology exhausts that of the open variety.

Lemma 4.1. *A smooth toric variety admits a smooth complete toric compactification $X \subset \overline{X}$ for which $\overline{X} \setminus X$ is the support of an effective Cartier divisor.*

Proof. Choose a complete rational fan containing Σ [8, Theorem 5.3 and Corollary 5.4]. It has a smooth refinement by star subdivisions which preserves every smooth cone [2, Theorem 11.1.9(a)–(c)]; the resulting smooth complete fan therefore still contains Σ.

Call a cone outside Σ bad if all its rays belong to Σ. If a bad cone exists, choose an inclusion-minimal one and star-subdivide at the sum of its primitive ray generators. Every proper face of this cone belongs to Σ, and the new ray lies outside the support of Σ. The subdivision preserves Σ and smoothness. Every new cone which was not already present contains the new ray, so no new bad cone is created, while the chosen bad cone is removed. The finite number of bad cones therefore strictly decreases. After finitely many steps, every cone outside Σ contains a ray outside Σ. Thus the complement is the union of the corresponding invariant divisors. Their reduced sum is Cartier because $\overline{X}$ is smooth. □

Proof of Theorem 1.1. Assume the cohomological condition and choose a compactification as in Lemma 4.1, with effective Cartier boundary E, and extend the invariant divisor D to $\overline{D}$ on $\overline{X}$. For any complete linear inclusion $\overline{Y} \hookrightarrow \overline{X}$, its intersection $Y = \overline{Y} \cap X$ is a linear inclusion in X. The hypothesis gives

$$0 \neq R\Gamma(Y, \mathcal{O}_X(D)|_Y) \simeq \operatorname*{colim}_{a \geq 0} R\Gamma(\overline{Y}, \mathcal{O}_{\overline{X}}(\overline{D} + aE)|_{\overline{Y}}).$$

The identity follows from localisation along the Cartier divisor and cohomology commuting with filtered colimits on a quasi-compact separated scheme [9, Section 30.6]. A nonzero class in this colimit remains nonzero at every sufficiently late stage.

There are only finitely many complete linear inclusions in $\overline{X}$; their subspaces are spanned by rays of the complete restricted fan. Choose one a making all their restriction cohomologies nonzero. Proposition 3.2 applies to $\overline{D} + aE$. Restricting its generators to X gives $F_{\ell *}\mathcal{O}_X(D)$, because finite direct image commutes with this open restriction. Restriction takes a generator of $\mathrm{Perf}(\overline{X})$ to a generator of $\mathrm{Perf}(X)$; every coherent sheaf on X extends coherently to $\overline{X}$ [9, Tag 01PF], and the extension is perfect because $\overline{X}$ is smooth. Thus generation holds for every sufficiently large exponent, completing the proof. □

5. **A finite criterion in the fan**

The geometric criterion ranges over rational subspaces, but two reductions make it finite; extra directions without rays contribute only torus factors, and the cohomology of each remaining restriction is determined by which ray inequalities fail.

Only subspaces spanned by rays need to be tested. Indeed, if W is linear and U is the span of the rays in $\Sigma|W$, then $\sigma \cap U = \sigma \cap W$ for every $\sigma \in \Sigma$. The saturated inclusion $N \cap U \subset N \cap W$ splits, giving

$$Y_W \simeq Y_U \times (\mathbb{G}_m)^{\dim W - \dim U}.$$

The toric divisor sequence shows that the restricted line bundle is pulled back from Y_U; the ray coordinates are unchanged and restriction of dual lattices is surjective. Its cohomology is therefore the cohomology on Y_U tensored with a Laurent polynomial ring, which leaves the vanishing condition unchanged.

It therefore suffices to enumerate the distinct spans U of subsets of $\Sigma(1)$, including $U = 0$, and retain those for which every $\sigma \cap U$ is a cone of Σ; this requires only a finite rational polyhedral calculation. For each retained U, put $M_U = (N \cap U)^\vee$ and let Δ_U be the simplicial complex of ray sets of cones in $\Sigma|U$. Write $D = \sum b_\rho D_\rho$. Its restriction to Y_U has these same coefficients on the rays contained in U; the other invariant divisors are disjoint from Y_U.

Lemma 5.1. *For $m \in M_U$, set $I_D(m) = \{\rho : \langle m, v_\rho\rangle + b_\rho < 0\}$. Then*

$$H^p(Y_U, \mathcal{O}(D)|_{Y_U})_m \simeq \widetilde{H}^{p-1}(\Delta_U[I_D(m)]; k), \qquad p \geq 0. \tag{7}$$

The formula holds for incomplete fans, with $\widetilde{H}^{-1}(\{\varnothing\}; k) = k$.

Proof. Cover Y_U by its maximal affine toric charts, indexed by the vertices of a simplex P. A weight-m section on an intersection chart is a copy of k exactly when that cone contains no ray in $I_D(m)$; all nonzero restriction maps are identities in the rational-monomial basis. Thus the weight-m Čech complex is $C^*(P, N_I; k)$, where N_I consists of the chart-index sets whose intersection contains a ray of $I = I_D(m)$. Since P is contractible, its cohomology is $\widetilde{H}^{p-1}(N_I; k)$.

Cover $\Delta_U[I]$ by its intersections with the maximal cones. Each nonempty intersection is a simplex, and the nerve is N_I. The nerve cohomology calculation gives (7), including the case $I = \varnothing$ through augmented cochains. □

For each retained U, compute

$$\mathcal{T}_U = \{I \subset \Sigma_U(1) : \widetilde{H}^j(\Delta_U[I]; k) \neq 0 \text{ for some } j \geq -1\}.$$

For $I \in \mathcal{T}_U$, let

$$P_{U,I}(b) = \left\{ m \in (M_U)_{\mathbb{R}} : \begin{array}{ll} \langle m, v_\rho \rangle + b_\rho \le -1, & \rho \in I, \\ \langle m, v_\rho \rangle + b_\rho \ge 0, & \rho \notin I \end{array} \right\}.$$

Theorem 5.2. *A toric power-map direct image of $\mathcal{O}_X(D)$ generates if and only if, for every retained ray span U, at least one of the polyhedra $P_{U,I}(b)$, $I \in \mathcal{T}_U$, contains a lattice point.*

Proof. Torus-weight decomposition and Lemma 5.1 identify this condition with nonzero cohomology on every Y_U. The torus-factor reduction then gives precisely the linear tests of Theorem 1.1. □

The two parts of the test are effective; reduced cohomology is finite linear algebra over $\mathbb{Q}$, and Cooper's elimination procedure decides integer feasibility and returns witnesses [1]. Leaving the coefficients b_ρ as free integer variables produces a finite Boolean combination of linear inequalities and congruences describing all successful divisors on a fixed fan. Replacing b by a principal translate translates the variables m, so the criterion depends only on the divisor class.

6. **One successful exponent**

Having decided whether generation is possible, we can produce an exponent by identifying a finite set of summands that occurs at all sufficiently large multiples of a single denominator. We combine a rational grid meeting every cell of the ray arrangement with the finitely many sign patterns of integral translates of the divisor.

Let $A : M \to \mathbb{Z}^s$ be the ray-pairing matrix in lattice bases, so $(Am)_i = \langle m, v_i \rangle$, and put $P = \operatorname{coker} A = \operatorname{Pic}(X)$. We retain the full group P, including its torsion. Write

$$\pi : \mathbb{R}^s \to P_{\mathbb{R}}, \qquad Z = \pi([-1, 0]^s).$$

The distinct line-bundle summands of $F_{\ell *}\mathcal{O}_X(D)$ are

$$\operatorname{Frob}_\ell(b) = \left\{ \left[\left\lfloor \frac{b + Au}{\ell} \right\rfloor \right] : u \in M/\ell M \right\} \subset P. \tag{8}$$

The floor is coordinatewise. To obtain this splitting, decompose the monomial sections on each affine chart by their classes modulo ℓM; the section inequalities in the class u are those of the divisor displayed in (8). The decompositions agree on chart intersections.

Append the coordinate rows to A. Let Δ be the least common multiple of its nonzero $n \times n$ minors, and set

$$R = \Delta \operatorname{lcm}(1, \dots, n + 1).$$

The identity minor makes Δ positive. If $n = 0$, take $\Delta = R = 1$.

The coefficients 0 and -1 record the two sides of an integral hyperplane, so the following rounding rules differ only when a ray pairing is integral. We retain both to include the cells lying on the hyperplanes.

For $e \in \{0, -1\}^s$, define

$$\Phi_e(x)_i = \begin{cases} \lfloor \langle x, v_i \rangle \rfloor, & e_i = 0, \\ \lceil \langle x, v_i \rangle \rceil - 1, & e_i = -1. \end{cases}$$

Lemma 6.1. *The range Θ_e of Φ_e in the divisor class group satisfies*

$$\Theta_e = \mathrm{Frob}_R(e) = \bigcup_{\ell \geq 1} \mathrm{Frob}_\ell(e).$$

Moreover, every class $a \in P$ with $a_{\mathbb{R}} \in Z$ has an integral representative, still denoted a, and $r_a \in R^{-1}M$ such that $a - Ar_a \in [-1, 0]^s$.

Proof. Cut the integral ray-hyperplane arrangement by the unit cube. Cramer's rule puts all cell vertices in $\Delta^{-1}M$. The average of $h + 1$ affinely independent vertices of an h-dimensional cell lies in its relative interior and in $R^{-1}M$. Thus the R-grid meets every relatively open cell, on which every Φ_e is constant. At $x = u/\ell$, its two rounding rules are exactly those in (8) with coefficients 0 and -1.

For the last assertion, choose an integral representative a and x with $a - Ax \in [-1, 0]^s$. Choose the first rounding rule when the difference is 0, the second when it is -1, and either elsewhere. Then $a = \Phi_e(x)$. Translate into the unit cube and replace x by an R-grid point in the same cell. This changes no divisor class and proves the assertion, including its torsion component. □

Let $\mathcal{H}$ contain both orientations of every primitive support-minimal integral relation $h \in \ker A^t \setminus \{0\}$. Set

$$c_h = \sum_i \max(-h_i, 0), \qquad C_b = 1 + \max\big(\{0\} \cup \{h \cdot b - c_h : h \in \mathcal{H}\}\big).$$

These relations are the facet normals of Z. Indeed, the face maximising h is generated by coordinate directions on which $h_i = 0$; its annihilator in $\ker A^t$ has dimension one precisely when h is support-minimal. Its maximum is c_h. Every summand class in (8) belongs to $\ell^{-1}[b] + (1 - \ell^{-1})Z$. Since $h(a)$ is integral,

$$h(a) \leq c_h + \frac{h(b) - c_h}{\ell} < c_h + 1 \quad (\ell \geq C_b)$$

implies $a_{\mathbb{R}} \in Z$.

Call $e \in \{0,-1\}^s$ *accessible* if an integer $w_e \in M$ satisfies

$$(b + Aw_e)_i \geq 0 \quad (e_i = 0), \qquad (b + Aw_e)_i \leq -1 \quad (e_i = -1).$$

Choose one witness for each accessible pattern and put

$$K_b = 1 + \max_{e,i} |(b + Aw_e)_i|, \qquad \mathcal{S}_b = \bigcup_{e \text{ accessible}} \Theta_e.$$

There is at least one accessible pattern. For $s = 0$ set $K_b = 1$.

An accessible pattern is a divisor representative that appears after one sufficiently large pushforward; a second pushforward, at the fixed exponent R, recovers its entire cell pattern Θ_e. The proposition shows that these patterns exhaust the summands, so their finite union suffices to test generation.

Proposition 6.2. *If $k \geq K_b$ and $Rk \geq C_b$, then $\mathrm{Frob}_{Rk}(b) = \mathcal{S}_b$. Consequently*

$$\ell_* = R \max(K_b, C_b)$$

has the property that $(F_{\ell_*})_* \mathcal{O}_X(D)$ generates exactly when a direct image under some power map generates.

Proof. For $k \geq K_b$, the floor of $(b + Aw_e)/k$ is e. Thus $\mathcal{O}_X(e)$ is a summand of $F_{k*}\mathcal{O}_X(D)$; applying F_{R*} and Lemma 6.1 gives $\mathcal{S}_b \subset \mathrm{Frob}_{Rk}(b)$.

Conversely, let $\ell = Rk \geq C_b$ and let a be a summand class. Choose a, r_a as in Lemma 6.1, absorbing a change $a \mapsto a + Am$ by $u \mapsto u + \ell m$, and a representative $u \in M$ for which $c = b + Au - \ell a \in \{0, \ldots, \ell - 1\}^s$. Since $\ell r_a \in M$, the vector $w = u - \ell r_a$ is integral and

$$b + Aw = c + \ell(a - Ar_a).$$

Let e record its signs. If $(a - Ar_a)_i = 0$ the sign is nonnegative; if it is -1 the sign is negative; in the interior either rounding rule gives a_i. Hence $a = \Phi_e(r_a) \in \mathcal{S}_b$.

A power-map direct image takes classical generators to generators. Indeed, the projection formula and the split trace make every perfect E a summand of $F_{a*}F_a^*E = E \otimes F_{a*}\mathcal{O}_X$. If some exponent ℓ_0 generates, a sufficiently large common multiple of ℓ_0 and R therefore generates and has the same distinct summands $\mathcal{S}_b$ as exponent ℓ_*. Multiplicities do not affect the thick closure. □

The construction uses only determinants, primitive relations, and integer feasibility with returned witnesses, so the exponent can be computed before testing generation. Different coordinates or witnesses may change ℓ_* but preserve its conclusion.

7. **A universal period and a linear threshold**

To pass from multiples of R to every large exponent, retain the residue class of the exponent and examine the lattice-point problem defining each summand. Beyond a bound linear in the divisor coefficients, the real polytopes have fixed vertex data and expand in fixed rational directions. Clearing their denominators leaves a lattice polytope whose large dilates absorb all remaining lattice errors.

The next lemma isolates the last step; projection removes the directions in which the polytope expands, and once the projected congruence is satisfied, a bounded fundamental parallelepiped contains the entire error that must be absorbed.

Lemma 7.1. *Let V be a rational subspace of a lattice space $M_{\mathbb{R}}$, and let $K \subset V$ be a lattice polytope containing 0 and spanning V. There is an effectively computable integer $T_K \geq 0$ such that, for every subset $P \subset M_{\mathbb{R}}$ and every integer $t \geq T_K$,*

$$(P + tK) \cap M \neq \varnothing \quad \Longleftrightarrow \quad \mathrm{pr}_V(P) \cap \mathrm{pr}_V(M) \neq \varnothing.$$

Here pr_V is the quotient map modulo V.

Proof. Choose a closed fundamental parallelepiped $\mathcal{P}$ for $M \cap V$. Choose affinely independent vertices $z_0, \ldots, z_d$ of K, where $d = \dim V$, and put $w = \sum z_i$, $h = d + 1$. Then w/h is in the relative interior. A sufficiently large integer a therefore satisfies

$$aw - \mathcal{P} \subset ahK.$$

This can be found from the finitely many facet inequalities of K; take $T_K = ah$. If $p \in P$ has the same image as some $m \in M$, adjust m by $M \cap V$ so that $p - m \in \mathcal{P}$. Then $m + aw \in p + T_K K \subset p + tK$, since $0 \in K$. The reverse implication follows by projection. For $V = 0$ take $T_K = 0$. □

An explicit choice is useful. Write $K = \{z \in V : C_j z \leq k_j\}$, choose a lattice basis $w_1, \ldots, w_d$ of $M \cap V$, and set $\beta_j = hk_j - C_j w$. For $\beta_j > 0$, choose

$$a \geq \left\lceil \frac{\sum_i \max(0, -C_j w_i)}{\beta_j} \right\rceil.$$

If $\beta_j = 0$, the corresponding functional is constant on K, hence vanishes on V and has $k_j = 0$; it imposes no condition. The maximum of these integers and 0 gives the a used above.

Theorem 7.2. *Fix the ray matrix A. There are computable integers $H_A \geq 1$ and $T_A \geq 0$, depending only on A, such that the raw coefficient sets*

$$\mathcal{R}_\ell(b) = \left\{ \left\lfloor \frac{b + Au}{\ell} \right\rfloor : u \in \{0, \ldots, \ell - 1\}^n \right\}$$

satisfy

$$\mathcal{R}_{\ell+\Delta}(b) = \mathcal{R}_\ell(b) \quad \textit{for} \quad \ell \geq N(b) := (1 + \max_i |b_i|)H_A + 1 + \Delta T_A.$$

The maximum is understood to be 0 *when there are no rays. If the linear cohomology criterion holds, every* $\ell \geq N(b)$ *generates. Thus both the eventual period and the coefficient of the divisor height in the generation bound depend only on the rays.*

Proof. We first stabilize the real polytopes, then their lattice-point conditions in each residue class of the exponent. The case $n = 0$ is immediate, so assume $n > 0$. Put $B = 1 + \max_i |b_i|$, with $B = 1$ if $s = 0$, and $S_i = \sum_j |A_{ij}|$. For $\ell \geq B$, every raw coefficient vector lies in the fixed box

$$\mathcal{C}_A = \prod_i ([-S_i - 1, S_i + 1] \cap \mathbb{Z}).$$

For $c \in \mathcal{C}_A$, define

$$C = \begin{pmatrix} -I \\ I \\ -A \\ A \end{pmatrix}, \quad d_c = \begin{pmatrix} 0 \\ \mathbf{1} \\ -c \\ c + \mathbf{1} \end{pmatrix}, \quad e_b = \begin{pmatrix} 0 \\ -\mathbf{1} \\ b \\ -b - \mathbf{1} \end{pmatrix}.$$

Then c occurs precisely when $P_{c,\ell} = \{u : Cu \leq \ell d_c + e_b\}$ contains an integer point. The four blocks of inequalities express

$$0 \leq u \leq \ell - 1, \qquad \ell c \leq b + Au \leq \ell(c + \mathbf{1}) - \mathbf{1}.$$

Thus the polytope keeps track of actual coefficient vectors, before passing to divisor classes.

For each invertible n-row submatrix C_I, its candidate vertex is $u_I(\ell) = \ell x_I + y_I$. Set

$$H_A = \max_{I,j} \left(|\det C_I| + \sum_k |(C_j \operatorname{adj} C_I)_k| \right), \qquad L = BH_A + 1.$$

After clearing the determinant, every feasibility inequality for this vertex has integral slope and constant term of absolute value at most BH_A. Its sign is therefore constant for real $\ell \geq L$. The list of feasible vertex bases is constant on this interval.

Put $Q_c = \{u : Cu \leq d_c\} \subset [0, 1]^n$. If $P_{c,L}$ is empty, all later polytopes are empty. Otherwise, each feasible vertex has slope in Q_c, and

$$P_{c,\ell} = P_{c,L} + (\ell - L)Q_c \qquad (\ell \geq L). \tag{9}$$

The forward inclusion of the right side into the left follows from the inequalities; the reverse follows by expressing each vertex as $u_I(L) + (\ell - L)x_I$ and taking convex hulls. This also proves the assertion for lower-dimensional polytopes.

Every vertex of Q_c lies in $\Delta^{-1}M$. Choose one, q_c, and put $V_c = \operatorname{span}(Q_c - q_c)$ and $K_c = \Delta(Q_c - q_c)$. For $0 \le r < \Delta$, formula (9) gives

$$P_{c,L+r+t\Delta} = P_{c,L+r} + t\Delta q_c + tK_c.$$

The middle translation is integral. By Lemma 7.1, integer feasibility is independent of t once $t \ge T_{K_c}$. Take T_A to be the maximum of these finitely many bounds over nonempty Q_c. All its data depend only on A. For $\ell \ge L + \Delta T_A$, each residue class has reached this stable range, proving the periodicity.

Finally, the linear criterion gives generation at all sufficiently large exponents by Theorem 1.1. Each exponent at least $N(b)$ has exactly the same raw summands as some arbitrarily large exponent in its residue class. Its summands therefore generate as well. □

Appendix A. A cellular construction of the sheaf model

The diagonal and linear-subvariety resolutions of [6] give an alternative construction of the inputs used in Section 3, with explicit canonical-bundle corrections and incidence maps. Let X be smooth and complete of dimension n, and retain the real torus $T = M_{\mathbb{R}}/M$ and its ray arrangement. For the generation argument, a fully faithful model with the compact lift and the power-map and slice functorialities suffices; we construct these directly.

A.1. Labels and the adjunction

Dualizing a resolution of the diagonal produces $\Delta_*\omega_X^{-1}[-n]$. The labels below make the adjoint pairing contribute the compensating factor $\omega_X[n]$, so that the resulting integral transform has kernel $\mathcal{O}_\Delta$. Keeping this correction in the labels also makes the final linear-slice pairing unshifted.

For $x \in M_{\mathbb{R}}$, define

$$L(x) = \mathcal{O}_X\left(\sum_\rho \lfloor -\langle x, v_\rho\rangle \rfloor D_\rho\right),$$

$$P(x) = L(x) \otimes \omega_X^{-1}[-n], \qquad Q(x) = L(-x)^\vee.$$

If t is a face of s, the coefficient inequality gives a canonical monomial inclusion $L(s) \to L(t)$, using simultaneous lifts. Changing lifts gives the corresponding character trivialisation. The reflected dual labels Q therefore form a sheaf of line bundles on the torus arrangement.

Choose a finite regular triangulation $\mathcal{K}$ refining the arrangement, fine enough that every closed star lifts injectively. Its constructible sheaves are covariant diagrams on

the face poset. For a cell s, let $h_s(t) = k$ if $s \leq t$, and 0 otherwise. This representable is the extension by zero from the open star and satisfies $\mathrm{RHom}(h_s, F) = F(s)$. These objects compactly generate the unbounded diagram category, as can be seen from the finite bar resolution with terms

$$\bigoplus_{s_0 < \cdots < s_r} h_{s_r} \otimes F(s_0).$$

Evaluation at a cell gives the augmented simplicial replacement of its face category, which has a terminal object.

Define

$$A_{\mathrm{cell}}(F) = R\Gamma(T, Q \otimes_k F).$$

The right side is the finite cellular cochain complex with coefficients in $D_{\mathrm{qc}}(X)$; its differential is signed incidence followed by generisation. These usual cellular sheaf constructions [3] remain valid for unbounded coefficient complexes because the cellular direction has finite length. The functor preserves colimits and has a right adjoint B.

Lemma A.1. *For a point x in the cell s,*

$$A_{\mathrm{cell}}(h_s) \simeq P(x).$$

These identifications take the representable map $h_t \to h_s$, for $s \leq t$, to the canonical monomial map $P(t) \to P(s)$. Consequently

$$B(E)(x) = \mathrm{RHom}_X(P(x), E). \tag{10}$$

Proof. On an affine toric chart U_σ, the weight-m part of Q is the extension by zero of the constant sheaf on the open region

$$\Omega_{\sigma,m} = \{z : \langle z, v_\rho \rangle < \langle m, v_\rho \rangle + 1 \text{ for } \rho \in \sigma(1)\}.$$

Its boundary hyperplanes belong to the arrangement. If x is outside its closure, it misses the open star of s. Otherwise the intersection with that star is a nonempty contractible oriented open n-manifold; radial contraction towards x reduces it to the corresponding convex region in a small ball. A covector negative on the pointed cone σ gives nonemptiness at an active boundary. Its compact cohomology is therefore $k[-n]$ exactly when

$$\langle m, v_\rho \rangle \geq \lceil \langle x, v_\rho \rangle \rceil - 1 \quad (\rho \in \sigma(1)).$$

These are the section inequalities for $L(x) \otimes \omega_X^{-1}$.

Fixing the orientation identifies the nonzero compact top classes. Chart monomials, affine-chart localisation, and inclusions of open stars take these classes to

the corresponding classes. Thus the weight identifications respect both the module actions and the incidence maps, and glue to $A_{\mathrm{cell}}(h_s) \simeq P(x)$. Adjunction gives (10) and its generisation maps. □

Artificial faces in the triangulation have identity maps within each original stratum, so $B(E)$ is constructible for the original arrangement. Each $P(x)$ is perfect, so its stalks, and hence B, commute with colimits.

A.2. **The diagonal and full faithfulness**

Identify the kernel torus of the diagonal by $x \mapsto (-x, x)$. Its labels in the resolution of [6, Theorem 3.5] are

$$H(x) = L(-x) \boxtimes L(x).$$

The boundary maps are the two canonical monomial maps, multiplied by the signed cellular incidence. The resulting chain complex resolves $\mathcal{O}_\Delta$. Subdivision to $\mathcal{K}$ preserves this complex up to quasi-isomorphism; send each oriented cell to the sum of its oriented top-dimensional subdividing simplices. Interior faces cancel and external faces carry the original monomial maps. Filtering by the original skeleta reduces the comparison to subdivision of a ball relative to its boundary, tensored with its coefficient line bundle.

Dualise this finite complex of vector bundles. With chain degree d placed in cohomological degree $-d$, multiplication in degree d by $(-1)^{d(d+1)/2}$ converts the dual differential into the cellular cochain differential. Hence

$$R\Gamma_T\big(L(-x)^\vee \boxtimes L(x)^\vee\big) \simeq R\mathcal{H}om_{X\times X}(\mathcal{O}_\Delta, \mathcal{O}_{X\times X}) \simeq \Delta_* \omega_X^{-1}[-n].$$

Formula (10) makes $A_{\mathrm{cell}}B$ the integral transform with this kernel tensored by $p_2^*\omega_X[n]$. The kernel is therefore $\mathcal{O}_\Delta$, giving a natural isomorphism $A_{\mathrm{cell}}B \simeq \mathrm{id}$.

This makes the adjunction counit invertible. Indeed, transporting the comonad across the natural isomorphism makes its counit and comultiplication endomorphisms ϵ, δ of the identity with $\epsilon\delta = 1$. Natural endomorphisms of the identity commute, so also $\delta\epsilon = 1$. Thus B is fully faithful on the entire unbounded category.

For an affine theta object $j_{\sigma*}\mathcal{O}_{U_\sigma}$, formula (10) gives the periodised costandard open dual cone, with its shift by n; the weight-m condition is $\langle m + x, v_\rho\rangle > 0$ on the rays of σ. The microsupport upstairs lies in $\bigcup_{\tau\preceq\sigma} \tau^\perp \times (-\tau)$. After translation and periodisation it lies in Λ_Σ. Affine Čech resolutions, colimit generation on affine charts, and preservation of the closed microsupport condition under colimits give $B(E) \in \mathrm{Sh}_{\Lambda_\Sigma}(T)$ for every E.

A.3. **The compact lift**

Give the invariant line bundles their divisor equivariant structures, with rational monomial χ^m of weight m. Upstairs, take the weight-zero part of derived Hom

$$K_D(x) = \mathrm{RHom}_{T_X\text{-equiv}}(P(x), \mathcal{O}_X(D)).$$

Canonical monomial maps have weight zero, so these are constructible cellular data. For $m \in M$, the principal-divisor identification gives

$$K_D(x+m) \simeq \mathrm{RHom}_X(P(x), \mathcal{O}_X(D))_m.$$

Summing over the lifts yields $p_! K_D \simeq B(\mathcal{O}_X(D))$, compatibly with all incidence and loop maps.

Choose a bounded lift of each arrangement cell. Properness makes each of the finitely many complexes $\mathrm{RHom}_X(P(s), \mathcal{O}_X(D))$ bounded with finite-dimensional cohomology, hence with finitely many nonzero torus weights. Thus K_D is supported on finitely many translates of these bounded cells. It is finite constructible and compactly supported, so $p_! K_D = p_* K_D$.

A.4. **Power maps and linear slices**

The relative dualising line of F_ℓ is $\omega_{F_\ell} = \omega_X \otimes F_\ell^* \omega_X^{-1}$. Finite duality and the projection formula give

$$\mathrm{RHom}_X(P(x), F_\ell^* E) \simeq \mathrm{RHom}_X(F_{\ell *} L(x) \otimes \omega_X^{-1}[-n], E).$$

Monomial residues and the elementary identity $\lfloor (\lfloor a \rfloor + b)/\ell \rfloor = \lfloor (a+b)/\ell \rfloor$ for integral b give

$$F_{\ell *} L(x) \simeq \bigoplus_{u \in M/\ell M} L\left(\frac{x-u}{\ell}\right).$$

Local monomial incidences preserve the residue class, while a loop transition $x \mapsto x+a$ replaces it by $u+a$ and leaves $(x-u)/\ell$ unchanged. The decomposition therefore respects the sheaf maps and gives

$$B(F_\ell^* E) \simeq f_{\ell !} B(E) \tag{11}$$

for arbitrary $E \in D_{\mathrm{qc}}(X)$.

For a complete linear inclusion $i : Y \hookrightarrow X$, put $V = N^\perp_{Y,\mathbb{R}}$ and $c = \dim V$. The subtorus T_V is a union of arrangement strata. The resolution of $i_* \mathcal{O}_Y$ in [6], with reflected coordinate $-x$, has labels $L(-x)$. The same subdivision and dualisation argument gives

$$A_{\mathrm{cell}}(k_{T_V}) = R\Gamma(T_V, Q|_{T_V}) \simeq R\mathcal{H}om_X(i_* \mathcal{O}_Y, \mathcal{O}_X) \simeq i_* \mathcal{O}_Y[-c].$$

The last equality uses the trivial normal bundle established in Lemma 2.1. Adjunction now yields

$$\mathrm{RHom}_T(k_{T_V}, B(\mathcal{O}_X(D))) \simeq R\Gamma(Y, \mathcal{O}_X(D)|_Y).$$

The source shift $[c]$ cancels the $[-c]$ in $i^!\mathcal{O}_X(D)$. This is precisely the unshifted pairing used in (3). Full faithfulness, (11), and the compact lift therefore supply all the sheaf-model inputs in the proof of Proposition 3.2.

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